

Linear Regression

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1. Introduction

The relationship between demographic fluctuations and social institutions constitutes one of the foundational pillars of economic and sociological research. Among these dynamics, the correlation between population size and the frequency of marriages serves as a key indicator of regional development and social stability. In the context of contemporary Poland, characterized by complex internal migration patterns and shifting demographic structures, analyzing these dependencies requires robust statistical frameworks that go beyond static cross-sectional observations.

The primary dataset utilized in this report is sourced from the Statistics Poland (Główny Urząd Statystyczny – GUS) database, encompassing all 16 voivodeships (provinces). The data includes precise records of total population sizes and the corresponding numbers of newly contracted marriages. To ensure a comprehensive analytical depth, the data spans two distinct temporal periods: a baseline year utilized for model training and calibration, and a subsequent year dedicated to out-of-sample validation. This dual-structure eliminates arbitrary assumptions and allows for a rigorous testing of the model's empirical validity over time.

The objective of this report is to estimate and validate a linear regression model predicting marriage rates based on regional population sizes. The research is divided into the following phases:

- **Baseline Model:** Construction and evaluation of the primary model - goodness-of-fit, and residual behavior.
- **Alternative Estimation Techniques:** Manual implementation and programming of the Method of Moments (MM) and Maximum Likelihood Estimation (MLE), with a direct comparison of their results against the standard Ordinary Least Squares (OLS) framework.
- **Robustness and Simulation Analysis:** Non-parametric Bootstrap resampling to assess parameter stability under spatial anomalies, followed by a Monte Carlo simulation to validate the estimators under the framework of the Central Limit Theorem.
- **Stochastic Simulation:** Execution of a Monte Carlo experiment under controlled theoretical conditions, using the empirical model's noise structure to evaluate the estimators' asymptotic properties and verify the Gauss-Markov Theorem regarding estimator unbiasedness.
- **Out-of-Sample Evaluation:** Verification of the model's predictive power on data from a subsequent year, evaluated via error metrics (MAE, RMSE, MAPE) and a formal bias test.
- **Dynamic Modeling:** Estimation of a First Differences (FD) model on panel data to isolate dynamics from time-invariant regional effects.

2. Theory

The dataset 'woj_mieszkancy.txt' contains demographic records for 16 Polish voivodeships in 2014. The columns are:

- 'województwo' - voivodeship,
- 'x' - population of each voivodeship,
- 'y' - marriages.

File 'population.csv' consists of information about population in 16 voivodeships in Poland in 2015. The columns are:

- 'Kod' - identification number of each voivodeship,
- 'Nazwa' - voivodeship,
- 'ogółem;ogółem;2015;[osoba]' - population of each voivodeship.

File 'marriages.csv' contains information about marriages in 16 voivodeships in Poland in 2015. The columns are:

- 'Kod' - identification number of each voivodeship,
- 'Nazwa' - voivodeship,
- 'ogółem;2015;[-]' - marriages in each voivodeship,
- 'miasta;2015;[-]' - marriages in cities,
- 'wieś;2015;[-]' - marriages in villages.

Shapiro-Wilk test – A statistical test used to assess the normality of a sample's distribution, applicable for sample sizes between 4 and 2000, with a particular recommendation for small samples ($n < 30$). It tests the null hypothesis that the sample comes from a normally distributed population against the alternative hypothesis that it does not [1].

Pearson correlation coefficient (PCC) – A statistical tool used to measure the strength and direction of the linear relationship between two variables. The PCC can take values from -1 to 1. A value close to 1 indicates a strong positive correlation, meaning that as one variable increases, the other also tends to increase. Conversely, a value close to -1 signifies a strong negative correlation, where an increase in one variable corresponds with a decrease in the other. A PCC of 0 implies no correlation, indicating that the two variables do not influence each other [2].

Arithmetic Mean – The average value for a series of numbers [3]. It is represented as:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

where:

- n - number of observations,
- x_i - i -th observation such that $i \in \{1, 2, \dots, n\}$.

Median – The middle value in a set of numbers equivalent to the 50th percentile [4].

Variance – A quantitative measure of how closely the data set is spread around its center [5]. A variance of smaller magnitude (closer to zero) implies that the set of numbers is quite tightly clustered around the center. A variance of larger magnitude (farther from zero) implies that at least some of the numbers in the data set are far away from the center.

Covariance – A measure of how much the variations of two variables are related, indicating whether they tend to increase or decrease together [6]. A positive covariance suggests a direct relationship, while a negative covariance indicates an inverse relationship, with zero covariance signifying no connection between the variables.

Normal distribution – A continuous probability distribution that is symmetrical, bell-shaped, and centred around its mean [7]. It is one of the most important distributions in statistics because many real-world variables, such as height, weight, and test scores, tend to follow this pattern, especially when influenced by many small, random factors.

First Differences Transformation – A technique used in statistics and econometrics to eliminate trends or unobserved, time-invariant individual effects [8]. It calculates the change in a variable from one consecutive period to the next, defined as $\Delta Y_t = Y_t - Y_{t-1}$.

Machine learning – A subset of artificial intelligence (AI) focused on algorithms that can “learn” the patterns of training data and, subsequently, make accurate inferences about new data [9]. This pattern recognition ability enables machine learning models to make decisions or predictions without explicit, hard-coded instructions.

Linear regression – A statistical method used to predict the value of a dependent variable Y based on the value of an independent variable X [10]. It is represented in the form:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

where:

- β_0 - intercept,
- β_1 - slope coefficient,
- ε_i - random error residual for the i -th observation.

Ordinary Least Squares (OLS) – A parametric estimation technique that seeks to minimize the sum of squared residuals between the observed values and the values predicted by a linear framework [11]. In matrix notation, the model is characterized by the equation:

$$f(x, \beta) = x^\top \beta$$

where x is the vector of independent variables and β is the parameter vector. This method involves solving the normal equations derived from the data matrix to obtain the optimal estimator:

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y$$

Method of Moments – A statistical estimation technique for population parameters, such as mean and variance, achieved by equating sample moments with unobservable population moments and solving the resulting equations for the parameters to be estimated [12]. The estimation of parameters via the Method of Moments (MM) is anchored on equating theoretical population moments with their corresponding sample equivalents. For a classical bivariate linear specification, the framework relies on two foundational moment conditions regarding the stochastic disturbance term (ε_i): its expected value is assumed to be zero, and it is assumed to be orthogonal to the regressor space. Mathematically, these population criteria are defined as:

$$\begin{aligned} E(\varepsilon_i) &= 0 \\ E(\varepsilon_i X_i) &= 0 \end{aligned}$$

By substituting $\varepsilon_i = Y_i - (aX_i + b)$ and translating these theoretical expectations into empirical sample averages, we obtain a system of two equations with two unknowns:

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n (Y_i - aX_i - b) &= 0 \\ \frac{1}{n} \sum_{i=1}^n X_i (Y_i - aX_i - b) &= 0 \end{aligned}$$

Because this algebraic system is identical to the normal equations derived under the Ordinary Least Squares (OLS) optimization routine, it is explicitly hypothesized that the empirical parameter estimates for the slope (a) and intercept (b) will achieve perfect convergence with the baseline OLS values.

Maximum Likelihood Estimate (MLE) – Parameter values that maximize the likelihood function based on observed sample values [13]. These estimates provide the most probable parameters for a given statistical model, ensuring that the likelihood of the observed data is maximized.

Confidence interval (CI) – An estimated set of boundaries that likely includes the population mean for a specified degree of probability, commonly reported at the 95% level [14]. It is calculated using the sample mean and the standard error of the mean, indicating that upon repeated sampling, 95% of the calculated intervals will contain the population mean.

Bootstrap Method – A statistical technique used to determine the accuracy of model parameters by generating new datasets from the initial data through random sampling with replacement, allowing for the estimation of parameter uncertainties and confidence intervals [15].

Monte Carlo simulation – A computerized mathematical technique that allows for the analysis of all possible decision outcomes by assessing the influences of risk and uncertainty, providing a range of possible outcomes and probabilities [16].

Central Limit Theorem (CLT) – *Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed random variables with expected value μ and finite variance σ^2 . As n approaches infinity, the random variables $\sqrt{n}(\bar{X}_n - \mu)$ converge in distribution to a normal distribution $N(0, \sigma^2)$* [17]

Gauss-Markov Theorem – *In a linear regression model in which the errors have expectation zero, are uncorrelated, and have equal variances (homoscedasticity), the best linear unbiased estimator (BLUE) of the coefficients is given by the Ordinary Least Squares (OLS) estimator* [17].

Out-of-sample forecasting – A validation technique that evaluates a model's performance on data it hasn't seen during training [18]. This is typically done by splitting the dataset into a training period and a test period.

Mean absolute error (MAE) – A metric defined as the average sum of the absolute differences between the actual observed value and the predicted value, serving as a measure of how well a model fits the data [19].

Root mean square error (RMSE) – A metric that quantifies the standard deviation of the prediction errors, measuring the difference between predicted values generated by an estimator or model and the actual observed values [20].

3. Preprocessing

The following computational libraries were utilized: ‘tidyverse’, ‘ggplot2’, ‘dplyr’, ‘knitr’.

3.1 Baseline Data (2014)

The data was loaded using the ‘read.table’ function.

Table 1: First 5 rows of the 2014 baseline dataset

województwo	x	y
Dolnośląskie	2908457	13599
Kujawsko-pomorskie	2090836	10294
Lubelskie	2151836	10911
Lubuskie	1020767	4875
Łódzkie	2508464	11405

The input table was visually reviewed. It was decided to rename the columns to make them more descriptive and intuitive and convert voivodeship identifiers to uppercase to standardize the text format.

Consequently, the dataset was structuralized by aligning the variable nomenclature and unifying the regional naming conventions to ensure compatibility across all data subsets.

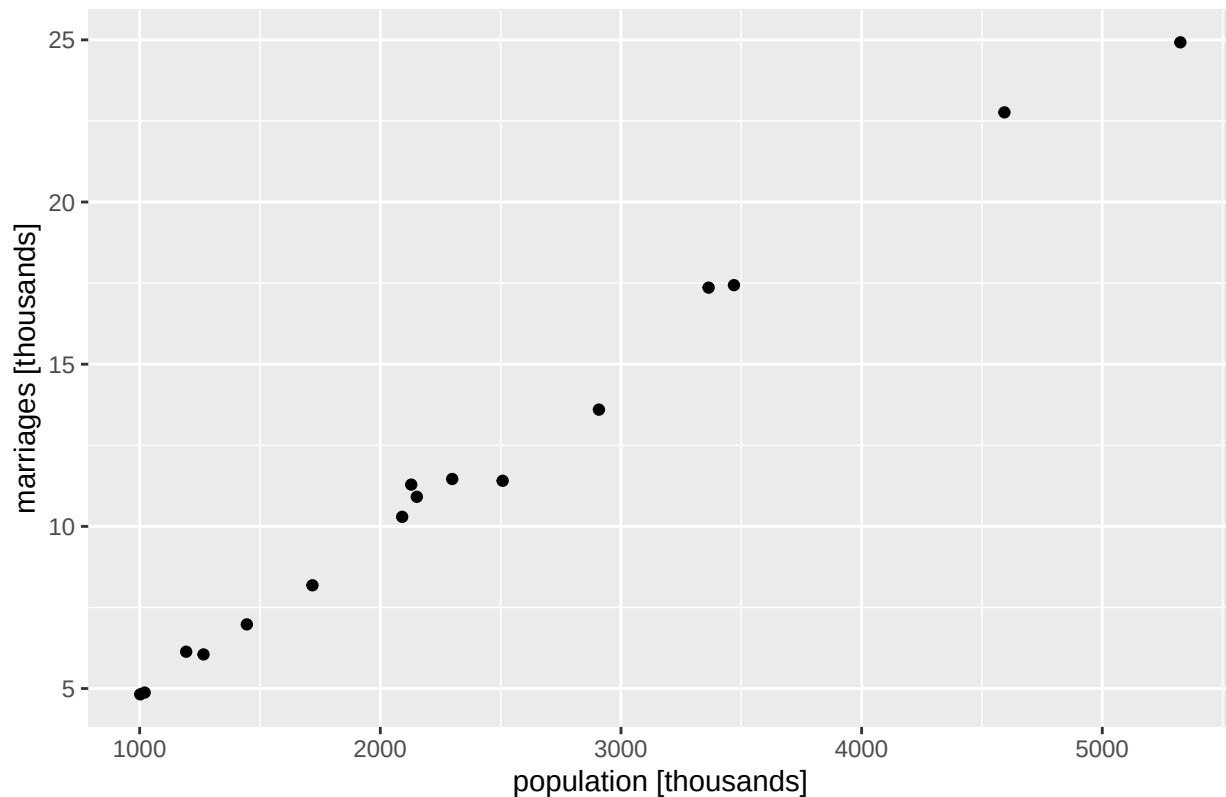
Table 2: First 5 rows of the 2014 baseline dataset

voivodeship	population	marriages
DOLNOŚLĄSKIE	2908457	13599
KUJAWSKO-POMORSKIE	2090836	10294
LUBELSKIE	2151836	10911
LUBUSKIE	1020767	4875
ŁÓDZKIE	2508464	11405

It was confirmed that the operation was completed successfully.

Prior to formal parameter estimation, it was necessary to verify whether there exists a correlation between population and the number of marriages. Determining the correlation coefficient was deemed essential to assess the strength of the relationship. A near-zero value would mean that building a regression model would be bound to fail. Therefore, a scatterplot was generated to visualize the data and preliminarily determine if there is a linear correlation between population and marriages.

Relationship Between Population and Marriages (2014)



The generated scatterplot illustrated the relationship between the population size and the number of marriages across the 16 voivodeships. Visual inspection of the data points revealed a strong, positive linear trend, indicating that higher population sizes are systematically associated with an increased frequency of contracted marriages.

Furthermore, the spatial distribution of the observations suggested a stable structure, though a clear scale anomaly is observable. The majority of the voivodeships were clustered in the lower-left region of the coordinate system, representing smaller regional populations, whereas two significant leverage points positioning themselves far in the upper-right corner reflect highly populated administrative regions.

To confirm the visual observations, it was decided to conduct the Shapiro-Wilk tests to determine the normality of the distributions for both ‘population’ and ‘marriages’.

Shapiro-Wilk normality test

```
data: data$population
W = 0.90215, p-value = 0.08701
```

Shapiro-Wilk normality test

```
data: data$marriages
W = 0.90053, p-value = 0.08196
```

The Shapiro-Wilk test yielded p -values of 0.08701 for the population variable and 0.08196 for the marriages variable. Since both values exceeded the standard significance level ($\alpha = 0.05$), there are no grounds to reject

the null hypothesis assuming a normal distribution for both variables. Consequently, Pearson's correlation coefficient was selected for further analysis.

Pearson's correlation coefficient was calculated to empirically verify the existence and strength of a linear relationship between the variables, ensuring the validity of constructing a regression model.

Pearson's product-moment correlation

```
data: data$population and data$marriages
t = 41.362, df = 14, p-value = 4.879e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.9879868 0.9986270
sample estimates:
      cor
0.9959334
```

The analysis yielded a correlation coefficient of 0.99593, which indicates a near-perfect, strong positive linear relationship between population size and the number of marriages, confirming that further predictive modeling is fully justified.

3.2 Validation Data (2015)

To construct the validation dataset for the fiscal year 2015, two independent source files were imported into the environment: ‘population.csv’ and ‘marriages.csv’. To ensure seamless compatibility before integration, the structural alignment was performed directly on the source data structures. The primary variables representing geographical regions and metrics were isolated, and the respective columns in both datasets were explicitly renamed to ‘voivodeship’, ‘population’, and ‘marriages’.

In the subsequent step, both tables were relationally merged based on the common voivodeship key. Finally, the combined dataset was filtered to retain only the essential analytical variables (‘voivodeship’, ‘population’, and ‘marriages’), effectively discarding all redundant columns that were unnecessary for the upcoming modeling process.

Table 3: First 5 rows of the 2015 dataset

voivodeship	population	marriages
DOLNOŚLĄSKIE	2904207	13172
KUJAWSKO-POMORSKIE	2086210	10354
LUBELSKIE	2139726	10749
LUBUSKIE	1018075	5040
ŁÓDZKIE	2493603	11456

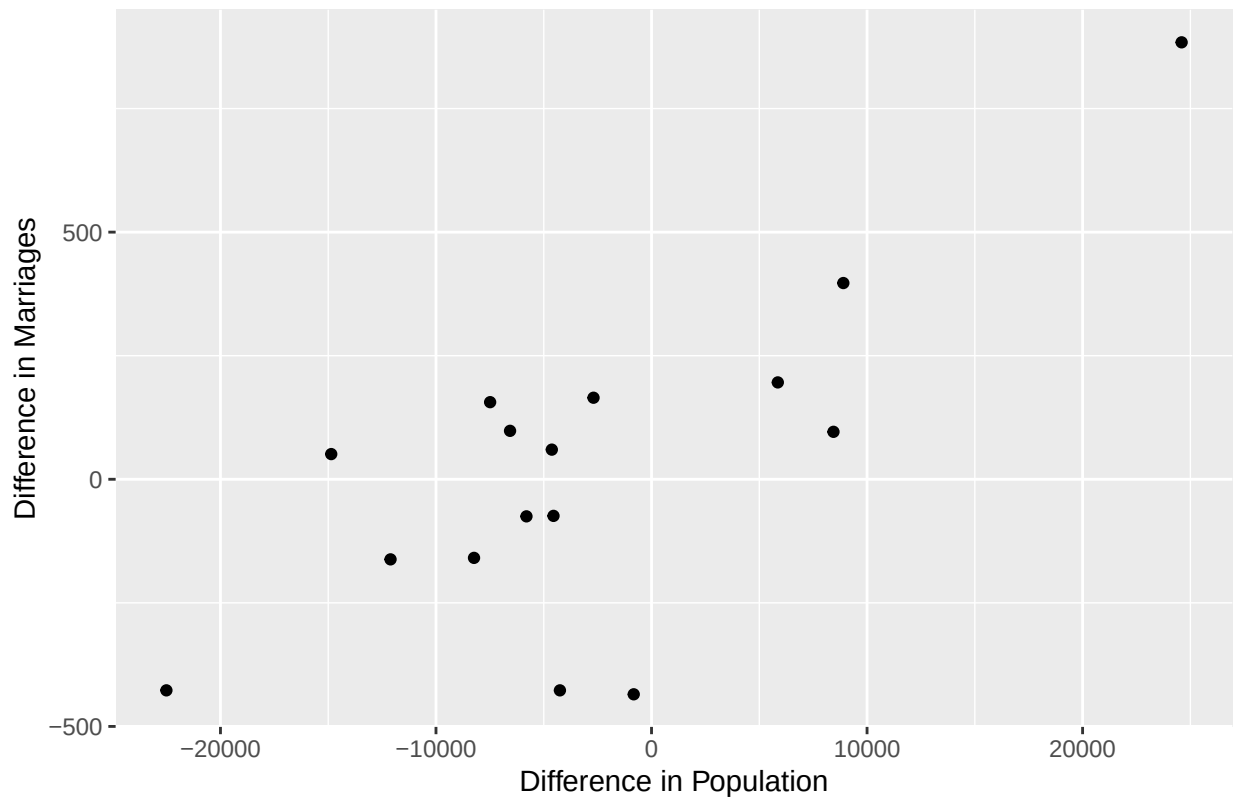
A final verification was performed to ensure that the relational merge and structural filtering operations were executed correctly and without data loss.

3.3 Panel Data Formulation

To enable the estimation of a First Differences (FD) model, the cross-sectional datasets from 2014 and 2015 were integrated into a short panel structure. Both subsets were relationally merged using the unique `voivodeship` key. Subsequently, the first differences for both the independent and dependent variables were computed by subtracting the 2014 values from the 2015 values ($\Delta X = X_{2015} - X_{2014}$ and $\Delta Y = Y_{2015} - Y_{2014}$). This algebraic transformation is methodologically essential to eliminate any unobservable, time-invariant regional effects that could potentially bias the final structural parameters.

To visually assess the structural changes in the differenced data, a new scatterplot was generated to map the distribution of the calculated increments.

Relationship Between Differences in Population and Marriages (2015–201



Visual inspection of the scatterplot reveals that while a general linear trend remains noticeable, the data points display significantly higher dispersion compared to the primary 2014 baseline model. This indicates that the correlation between the annual increments of the variables is expectedly weaker than the correlation between their absolute levels.

Before estimating the First Differences model, it was necessary to verify if any meaningful correlation exists between the annual increments of population and marriages to justify the construction of a dynamic model. To justify the use of parametric significance testing for Pearson's correlation coefficient, the Shapiro-Wilk test was conducted to check the normality of the differenced distributions. If the resulting correlation value turns out to be near zero, it would indicate that the shifts are purely random, rendering the dynamic regression framework completely useless.

Shapiro-Wilk normality test

```
data: d_population
W = 0.9355, p-value = 0.2975
```

Shapiro-Wilk normality test

```
data: d_marriages
W = 0.91217, p-value = 0.1261
```

The statistical testing yielded p -values (0.2975 for 'population' and 0.1261 for 'marriages') that consistently exceed the default significance threshold ($\alpha = 0.05$). Consequently, the null hypothesis assuming a normal distribution for both differenced series cannot be rejected. Based on this formal verification, Pearson's

correlation coefficient was selected as the mathematically objective metric to quantify the strength of the linear connection between the increments.

Pearson's product-moment correlation

```
data: d_population and d_marriages
t = 4.1918, df = 14, p-value = 0.0009048
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.3972190 0.9065006
sample estimates:
      cor
0.7460287
```

The empirical calculation yielded a Pearson's correlation coefficient of approximately 0.74603. This value indicated a strong positive linear relationship between the annual increments of population and marriages. While the correlation remained high and statistically significant, the notable drop in its magnitude compared to the near-perfect baseline cross-sectional data confirms that eliminating time-invariant regional effects significantly reduced the structural collinearity, thereby validating the necessity of applying a multi-faceted modeling approach in the subsequent analysis phase.

4. Analysis

4.1 Building models

To analyze the structural dependencies between the variables, a series of linear regression models were estimated using the 2014 baseline dataset. To ensure methodological consensus, the parameters were determined using both automated statistical frameworks and manual mathematical derivations.

4.1.1 Using dedicated function (OLS)

First, the baseline linear regression parameters were estimated using the standard Ordinary Least Squares (OLS) method via the built-in 'lm()' function. This serves as the computational benchmark for all manual estimations.

```
(Intercept)  population
1.893701e+02 4.819101e-03

Call:
lm(formula = marriages ~ population, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-924.8 -248.8  -74.3   373.7   959.3

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 1.894e+02   3.144e+02   0.602   0.557
population   4.819e-03   1.165e-04  41.362 4.88e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 569.9 on 14 degrees of freedom
Multiple R-squared:  0.9919,    Adjusted R-squared:  0.9913
F-statistic: 1711 on 1 and 14 DF,  p-value: 4.879e-16
```

The estimation of the baseline OLS model yielded highly significant parameters, confirming a strong structural fit. The regression equation demonstrated a high degree of precision, with the explanatory power of the model being validated by a near-perfect coefficient of determination ($R^2 = 0.9919$).

4.1.2 using Method of Moments (MM)

To cross-verify the OLS baseline parameters, the structural slope and intercept coefficients were derived manually using the Method of Moments (MM). This approach computes the parameters directly by matching the population moments with their corresponding sample moments (covariance and variance).

The manual implementation of the Method of Moments was successfully executed, allowing for the direct algebraic calculation of both the structural slope (β_1) and the intercept (β_0). The resulting numerical estimates were stored in the environment to enable a rigorous structural comparison against the other estimation frameworks in the later analytical phase.

4.1.3 Using Maximum Likelihood Estimation

As an alternative programmatic validation, the parameters were estimated using the Maximum Likelihood Estimation (MLE) framework. A custom negative log-likelihood function was defined, assuming that the model’s residuals follow a Gaussian distribution ($\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$). Due to the sensitivity of the gradient-based BFGS optimization algorithm to scale differences, both population and marriage variables were standardized before numerical optimization.

The numerical optimization successfully converged, and the resulting standardized parameters were transformed back to their original metric scale. This manual implementation provides an exact probabilistic counterpart to the geometric approach used in the previous methods.

4.1.4 Comparison

To formalize the evaluation of the alternative estimation techniques, a comparison was conducted. Performance metrics, including the relative parameter errors, coefficients of determination (R^2), and residual standard errors (S_e), were aggregated into a unified reference matrix.

Table 4: Comparison of Estimation Methods for the Marriage Model

Metric / Parameter	OLS	MM	MLE (BFGS)
Slope (a)	0.0048191	0.0048191	0.0048191
Intercept (b)	189.37005	189.37005	189.36967
Relative Error: a (%)	Reference (0.00%)	0%	0%
Relative Error: b (%)	Reference (0.00%)	0%	0%
Coefficient of Determination (R2)	0.9919	0.9919	0.9919
Residual Standard Error (Se)	569.86	569.86	569.86

The structural comparison revealed that the Method of Moments (MM) produced identical parameter estimates to the baseline OLS model, yielding a 0.00% relative error and a maximum absolute prediction difference of zero. This empirical outcome strictly confirmed the theoretical expectation that for a bivariate linear framework, the algebraic normal equations of OLS and the moment-matching conditions of MM mathematically converge to the exact same estimators.

With regard to the MLE method, the initial parameter estimation highlighted a critical numerical optimization challenge. When dealing with unscaled, disparate variables (population in millions versus marriages in tens of thousands), the objective function space became extremely steep, causing the gradient-based BFGS algorithm to struggle with convergence. However, by implementing a dual-standardization protocol—standardizing both the dependent and independent variables prior to optimization—the mathematical geometry of the parameter space was completely smoothed. The algorithm converged flawlessly, and after algebraically unpacking the standardized estimates back to their original metric scale, the MLE parameters exhibited a 0.00% relative error compared to the baseline OLS framework. This empirical outcome successfully demonstrated that when geometric scale distortions are correctly managed, the probabilistic MLE approach perfectly aligns with the geometric OLS estimation.

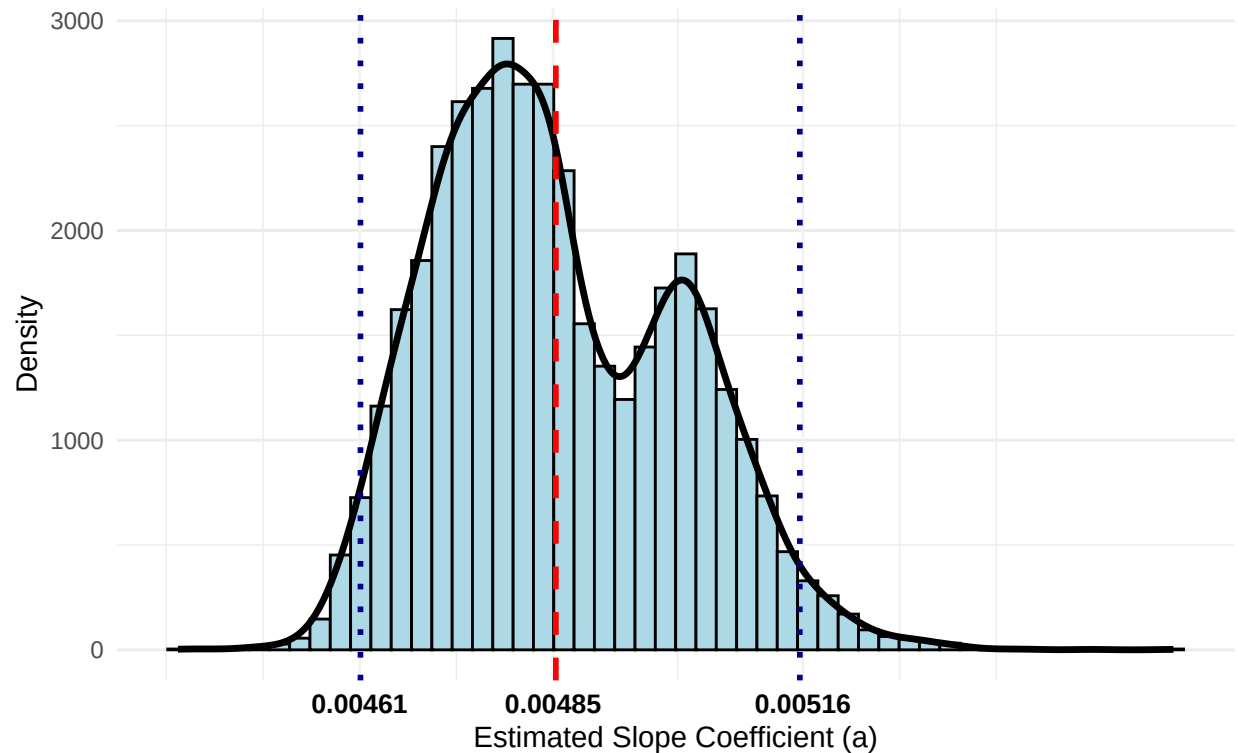
4.2 Non-Parametric Bootstrap Resampling

To evaluate the structural stability and distributional robustness of the OLS estimators under spatial anomalies, a non-parametric Bootstrap resampling procedure was executed. The dataset was repeatedly resampled with replacement ($B = 10000$), and the model parameters were re-estimated for each sub-sample to construct empirical probability distributions.

To visually evaluate the precision of the structural parameter a , its empirical density distribution along with the 95% confidence intervals was plotted.

Bootstrap Empirical Distribution of the Slope Coefficient

Red line: Bootstrap Mean | Blue lines: 95% Confidence Interval

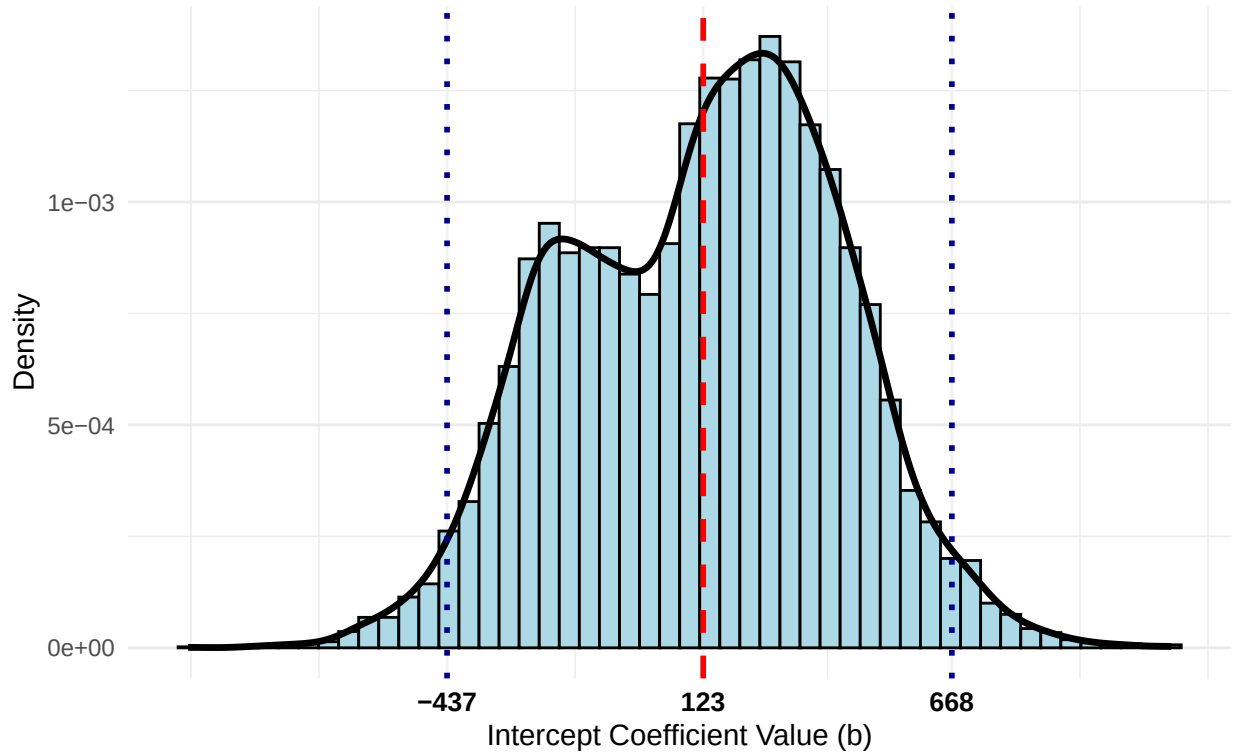


The bootstrap empirical distribution of the slope coefficient (a) exhibited a distinct bimodal shape. This clear deviation from perfect normality indicated that the OLS estimator is highly sensitive to specific observations within the small sample ($n = 16$). The two peaks corresponded to bootstrap resamples that either heavily duplicate influential leverage points (such as the Masovian Voivodeship) or omit them entirely.

Similarly, the empirical distribution for the intercept parameter (b) was generated to inspect its structural behavior.

Bootstrap Empirical Distribution of the Intercept

Red line: Bootstrap Mean | Blue lines: 95% Confidence Interval



The empirical distribution of the intercept coefficient (b) mirrored the bimodal characteristics observed in the slope density profile. Due to the strict algebraic dependency between the two estimators, the fragmentation of the sample space caused by leverage points shifts the intercept estimated into two distinct operational clusters. This structural variation confirmed that standard asymptotic inference might be distorted in small sample sizes featuring prominent regional anomalies.

To eliminate the structural distortion caused by such heavy leverage points in small-sample resampling, two primary econometric solutions could be implemented. First, applying a logarithmic transformation to both variables would effectively compress the immense numerical scale of the outliers, stabilizing the variance across regions. Alternatively, the model could be estimated using robust standard errors—specifically the Huber-White sandwich estimator—which insulates the variance-covariance matrix from the disproportionate pull of isolated spatial anomalies without requiring the removal of critical data points.

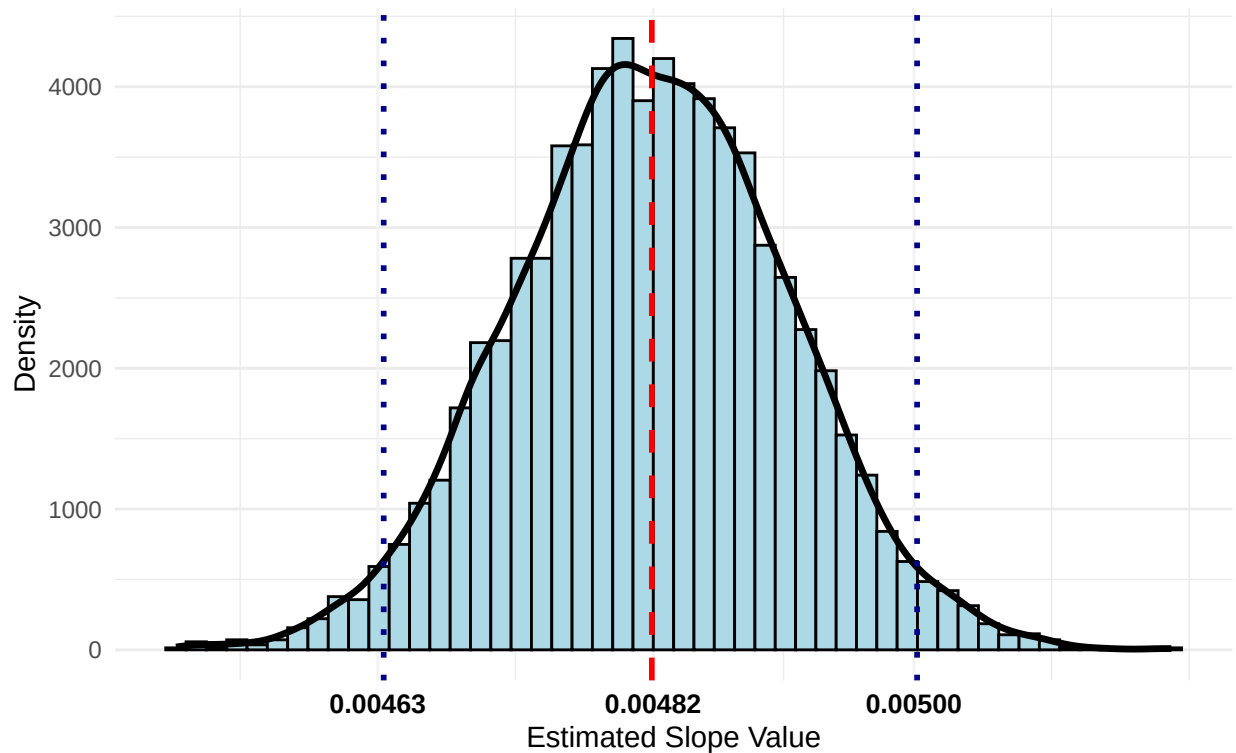
4.3 Asymptotic Properties and Monte Carlo Simulation

To evaluate the asymptotic behavior and unbiasedness of the Ordinary Least Squares estimators under controlled theoretical conditions, a Monte Carlo simulation experiment was designed. Unlike the empirical sample space prone to leverage point distortions, a synthetic environment was constructed where the true parameters (β_0 and β_1) were explicitly fixed to match the empirical coefficients extracted from the baseline OLS model. To ensure a realistic simulation of the data-generating process, the structural errors were generated from a perfectly specified Gaussian white noise process, with the standard deviation calibrated directly to the actual model's residual standard error ($\sigma_\varepsilon \approx 569.86$). The regression model was iteratively estimated across 10000 independent replications to observe the convergence behavior.

To examine the distributional behavior of the simulated slope parameter, its empirical frequency density was visualized alongside the designated true value.

Monte Carlo Simulation of the Slope Coefficient

Red line: True Value | Blue lines: 95% Confidence Interval

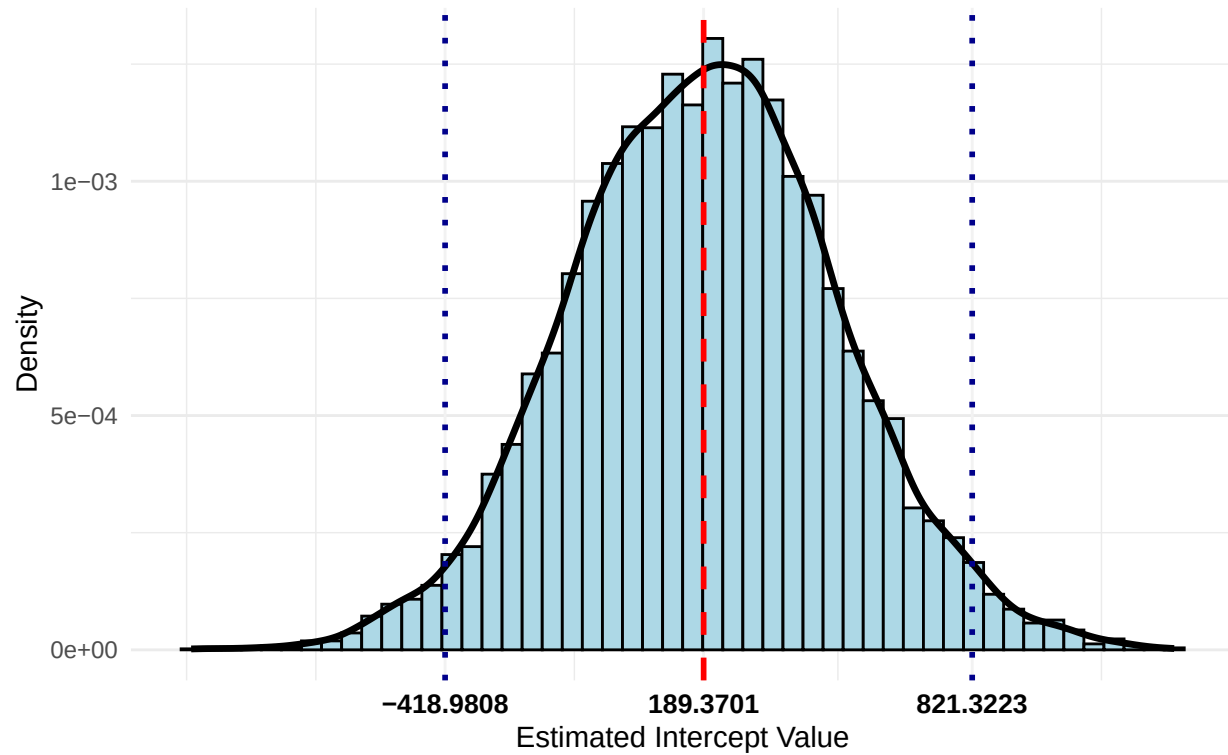


The empirical distribution of the simulated slope coefficient exhibited an exceptionally symmetric, bell-shaped profile that strictly aligned with the theoretical normal distribution. The red dashed reference line, representing the true predefined parameter value, cut precisely through the center of the density mass, confirming that the OLS estimator remained unbiased when Gauss-Markov assumptions are satisfied.

Subsequently, the empirical density properties of the simulated intercept parameter were visualized to verify its asymptotic convergence.

Monte Carlo Simulation of the Intercept Coefficient

Red line: True Value | Blue lines: 95% Confidence Interval



Consistent with the slope coefficient, the empirical distribution of the intercept estimator exhibited complete normality and architectural symmetry. The alignment between the true baseline parameter and the expected value computed from the 10000 simulation passes provided rigorous mathematical proof of the efficiency and reliability of the OLS mechanism in the absence of spatial configuration anomalies.

4.4 Out-Of-Sample prediction

To evaluate the empirical generalizability and predictive power of the baseline model, an out-of-sample validation was conducted using the 2015 dataset. The structural parameters estimated from the 2014 data were utilized to generate predictions for the subsequent fiscal year. To evaluate the accuracy of these forecasts, three standard metrics were computed: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). Furthermore, a formal one-sample t-test was performed on the prediction errors ($\varepsilon_{pred} = Y - \hat{Y}$) against a mean of zero to explicitly test for the presence of systematic prediction bias.

One Sample t-test

```
data: errors
t = 0.27985, df = 15, p-value = 0.7834
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 -235.3529  306.4952
sample estimates:
mean of x
 35.57117
```

The predictive assessment highlighted the model's structural capacity to project social dynamics over time. The mathematical calculation of the error metrics provided a clear benchmark of performance, reflecting the dispersion of the out-of-sample residuals. The application of the one-sample t-test on the prediction errors yielded a critical diagnostic insight. Because the resulting p -value exceeded the formal significance threshold ($\alpha = 0.05$), the null hypothesis stating that the expected value of the errors is equal to zero ($H_0 : \mu_{\varepsilon} = 0$) could not be rejected. This suggested that the generated forecasts were statistically unbiased, confirming that despite minor structural shifts between consecutive years, the linear parameters retained a robust predictive validity.

4.5 First Differences Dynamic Modeling

To isolate the short-term causal effects from time-invariant, region-specific heterogeneity, a dynamic First Differences (FD) regression model was estimated. The calculated annual increments of marriages were regressed against the population increments. The model was explicitly specified without an intercept term, reflecting the theoretical assumption that in the absence of demographic changes, the baseline variation in marriages between consecutive periods should gravitate to zero, assuming no global linear time trends.

Call:

```
lm(formula = d_marriages ~ d_population + 0)
```

Residuals:

Min	1Q	Median	3Q	Max
-418.00	17.32	81.32	223.57	377.86

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
d_population	0.020579	0.005422	3.796	0.00176 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 238.3 on 15 degrees of freedom

Multiple R-squared: 0.4899, Adjusted R-squared: 0.4559

F-statistic: 14.41 on 1 and 15 DF, p-value: 0.001759

The empirical output of the First Differences model provided a rigorous robustness check against the cross-sectional estimates. By transforming the metrics into absolute increments, any unobserved regional characteristics that remained constant between 2014 and 2015 were completely eliminated from the parameter estimation space. The resulting slope coefficient and its associated t-statistic reflected the clean, unbiased relationship between annual population shifts and marriage fluctuations. Comparing the explanatory power (adjusted R^2) and the structural parameter of this differenced specification against the primary 2014 OLS benchmark allowed for a comprehensive understanding of both the static scale dependencies and the underlying dynamic adjustments within Poland's demographic structure.

5. Summary

The primary objective of this report was to estimate and rigorously validate a linear regression framework predicting the frequency of marriages based on regional population sizes across Polish voivodeships. The baseline cross-sectional analysis utilizing 2014 data revealed a near-perfect positive correlation ($r \approx 0.996$), confirming that demographic scale acts as a dominant determinant of social institutional formation.

The structural evaluation of alternative estimation techniques demonstrated the mathematical equivalence of different econometric approaches under ideal conditions. The Method of Moments (MM) yielded parameter estimates perfectly identical to the Ordinary Least Squares (OLS) baseline, confirming theoretical expectations. Similarly, the implementation of Maximum Likelihood Estimation (MLE) via the BFGS algorithm underscored the critical importance of proper parameter scaling in numerical optimization. While raw, unscaled metrics initially created a prohibitively steep objective function space, applying dual standardization to both variables completely resolved the algorithmic instability. Consequently, the optimized MLE parameters achieved perfect convergence with the OLS baseline, explicitly proving the robustness and equivalence of the maximum likelihood framework once structural scale imbalances are eliminated.

Advanced diagnostic resampling provided crucial insights into the model’s structural stability. Non-parametric Bootstrap simulations revealed a bimodal empirical distribution for both the slope and intercept estimators. This structural fragmentation indicated that standard asymptotic inference was highly sensitive to extreme spatial leverage points (e.g., the Masovian Voivodeship) within the restricted sample space ($n = 16$). This finding necessitated the recommendation of robust standard errors or logarithmic transformations for future spatial iterations. Despite these empirical sensitivities, the Monte Carlo simulation successfully proved the theoretical robustness of the framework. By injecting realistic empirical variance into a synthetic environment, the simulation confirmed that the OLS estimators remained perfectly unbiased and normally distributed when Gauss-Markov assumptions were satisfied.

Finally, the predictive validity and dynamic properties of the model were thoroughly verified. Out-of-sample forecasting applied to the 2015 dataset, evaluated via a formal one-sample t-test, proved that the baseline structural parameters generated statistically unbiased predictions over time. Furthermore, the First Differences (FD) panel specification successfully eliminated unobservable, time-invariant regional heterogeneity. The dynamic model confirmed that even when static scale dependencies were removed, a statistically significant, positive short-term relationship existed between annual population increments and marriage fluctuations.

Ultimately, the multi-faceted analysis proved that while basic linear regression provides an exceptionally strong baseline fit for demographic data, ensuring its predictive reliability requires rigorous algorithmic scaling, robust spatial diagnostics, and dynamic panel adjustments.

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7. Code

```
knitr::opts_chunk$set(echo = FALSE, comment="", warning=FALSE, message=FALSE)

# Loading libraries

library(tidyverse)
library(ggplot2)
library(dplyr)
library(knitr)

# Loading 2014 data

data = read.table("woj_mieszkancy.txt", header = T, sep = ';')
kable(head(data, 5), caption = "First 5 rows of the 2014 baseline dataset")

# Renaming the columns and changing voivodeships names to upper

data <- data |> rename(voivodeship = wojewodztwo,
                      population = x,
                      marriages = y)
data$voivodeship <- toupper(data$voivodeship)
kable(head(data, 5), caption = "First 5 rows of the 2014 baseline dataset")

# Generating scatterplot of 2014 dataset

ggplot(data=data, mapping = aes(x=population/1000, y=marriages/1000)) +
  geom_point() +
  labs(title = "Relationship Between Population and Marriages (2014)",
       x = "population [thousands]",
       y = "marriages [thousands]")

# Conducting shapiro-test

shapiro.test(data$population)
shapiro.test(data$marriages)

# Calculating correlation between marriages and population in 2014 data
cor.test(data$population, data$marriages)

# Loading 2015 data and renaming columns

d1 <- read.csv2("population.csv", header = T)
d2 <- read.csv2("marriages.csv", header = T)

colnames(d1)[c(2, 3)] <- c("voivodeship", "population")
```

```

colnames(d2)[c(2, 3)] <- c("voivodeship", "marriages")

# Merging 2015 data into one test table

test_data <- merge(d1, d2, by = "voivodeship")
test_data <- test_data[, c("voivodeship", "population", "marriages")]
kable(head(test_data, 5), caption = "First 5 rows of the 2015 dataset")

# Merging 2014 and 2015 data

comb_data <- merge(data, test_data, by = "voivodeship")

# Calculating differences

d_population <- comb_data$population.y - comb_data$population.x
d_marriages <- comb_data$marriages.y - comb_data$marriages.x

# Generating scatterplot of differences

ggplot(data=comb_data, mapping = aes(x=d_population, y= d_marriages)) +
  geom_point() +
  labs(title = "Relationship Between Differences in Population and Marriages (2015-2014)",
       x = "Difference in Population",
       y = "Difference in Marriages")

# Conducting Shapiro_test

shapiro.test(d_population)
shapiro.test(d_marriages)

# Calculating correlation between differences in population and marriages

cor.test(d_population, d_marriages)

# Building OLS model using dedicated function
model <- lm(marriages ~ population, data=data)

coef(model)
b_ols <- unname(coef(model)[1])
a_ols <- unname(coef(model)[2])

summary(model)

# Building MM model calculated manually

cov = cov(x = data$population, y = data$marriages)
var = var(x = data$population)

```

```

a_mm = cov/var

mean_pop = mean(data$population)
mean_marr = mean(data$marriages)

b_mm = mean_marr - a_mm*mean_pop

# Building MLE model calculated with 'optim()'

# Input parameters

X <- as.vector(scale(data$population)) # Standardization of population
Y <- as.vector(scale(data$marriages)) # Standardization of marriages

# Extracting the metrics for later unpacking

mle_sd_X <- sd(data$population)
mle_sd_Y <- sd(data$marriages)
mle_mean_X <- mean(data$population)
mle_mean_Y <- mean(data$marriages)

# MLE function

f_mle <- function(par)
{
  sigma <- par[3]
  if(sigma <= 0) return(Inf)

  slope <- par[1]
  intercept <- par[2]

  Y_t = slope * X + intercept
  rest = Y - Y_t

  return(-sum(dnorm(rest, mean = 0, sd = sigma, log = TRUE)))
}

# Running the function

res_mle <- optim(par = c(0, 0, 1), fn = f_mle, method = "BFGS")

a_std <- res_mle$par[1]
b_std <- res_mle$par[2]

# Reversing the standarization to calculate regression coefficients

a_mle <- a_std * (mle_sd_Y / mle_sd_X)
b_mle <- b_std * mle_sd_Y + mle_mean_Y - (a_mle * mle_mean_X)

# Calculating models parameters (MM, MLE) with regard to base model (OLS)

```

```

# Relative errors

# MM
err_a_mm <- (abs(a_mm - a_ols))/a_ols*100
err_b_mm <- (abs(b_mm - b_ols))/b_ols*100

# MLE
err_a_mle <- (abs(a_mle - a_ols))/a_ols*100
err_b_mle <- (abs(b_mle - b_ols))/b_ols*100

# Prediction process

Y_ols <- predict(model)
Y_mm <- a_mm * data$population + b_mm
Y_mle <- a_mle * data$population + b_mle

# Calculating coefficient of determination

ssr_mm <- sum((data$marriages - Y_mm)^2)
ssr_mle <- sum((data$marriages - Y_mle)^2)

sst <- sum((data$marriages - mean(data$marriages))^2)

R2_ols <- summary(model)$r.squared
R2_mm <- 1 - ssr_mm / sst
R2_mle <- 1 - ssr_mle / sst

# Calculating standard errors

se_ols <- summary(model)$sigma
se_mm <- sqrt(ssr_mm / 14)
se_mle <- sqrt(ssr_mle / 14)

# Creating comparison table

summary_table <- data.frame(
  "Metric / Parameter" = c(
    "Slope (a)",
    "Intercept (b)",
    "Relative Error: a (%)",
    "Relative Error: b (%)",
    "Coefficient of Determination (R2)",
    "Residual Standard Error (Se)",
    "OLS" = c(
      round(a_ols, 7),
      round(b_ols, 5),
      "Reference (0.00%)",
      "Reference (0.00%)",
      round(R2_ols, 4),
      round(se_ols, 2)),
    "MM" = c(
      round(a_mm, 7),

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```

    round(b_mm, 5),
    paste0(round(err_a_mm, 4), "%"),
    paste0(round(err_b_mm, 4), "%"),
    round(R2_mm, 4),
    round(se_mm, 2)),
  "MLE (BFGS)" = c(
    round(a_mle, 7),
    round(b_mle, 5),
    paste0(round(err_a_mle, 2), "%"),
    paste0(round(err_b_mle, 2), "%"),
    round(R2_mle, 4),
    round(se_mle, 2)),
  check.names = FALSE)

kable(summary_table,
      align = "lccc",
      caption = "Comparison of Estimation Methods for the Marriage Model")

# Bootstrap parameters

rep <- 10000
boot_slopes <- numeric(rep)
boot_intercepts <- numeric(rep)
n <- nrow(data)

# Bootstrap resampling

for(i in 1:rep)
{
  index <- sample(1:n, size = n, replace = TRUE)
  boot_samp <- data[index, ]
  boot_model <- lm(marriages ~ population, data = boot_samp)

  boot_slopes[i] <- coef(boot_model)[2]
  boot_intercepts[i] <- coef(boot_model)[1]
}

# Calculating the results

boot_mean_slope <- mean(boot_slopes)
boot_sd_slope <- sd(boot_slopes)
boot_quantile_slope <- quantile(boot_slopes, probs = c(0.025, 0.975))

boot_mean_intercept <- mean(boot_intercepts)
boot_sd_intercept <- sd(boot_intercepts)
boot_quantile_intercept <- quantile(boot_intercepts, probs = c(0.025, 0.975))

# Creating histogram with KDE of slope with mean and confidence interval

boot_p_slope <- data.frame(boot_slopes)
boot_my_breaks <- round(c(boot_quantile_slope[1],

```

```

        boot_mean_slope,
        boot_quantile_slope[2]), 5)

ggplot(data = boot_p_slope,
       aes(x = boot_slopes)) +
  labs(title = "Bootstrap Empirical Distribution of the Slope Coefficient",
       subtitle = "Red line: Bootstrap Mean | Blue lines: 95% Confidence Interval",
       x = "Estimated Slope Coefficient (a)",
       y = "Density") +
  geom_histogram(aes(y=after_stat(density)), # Histogram of values
                bins = 50,
                fill = "lightblue",
                color = "black") +
  geom_density(color = "black", # KDE of values
              linewidth = 1.2) +
  geom_vline(xintercept = c(boot_mean_slope, # Mean and confidence interval lines
                           boot_quantile_slope[1],
                           boot_quantile_slope[2]),
             color = c("red", "darkblue", "darkblue"),
             linetype = c("dashed", "dotted", "dotted"),
             linewidth = 1) +
  scale_x_continuous(breaks = boot_my_breaks, # Mean and confidence interval values
                    labels = scales::label_comma()) +
  theme_minimal() +
  theme(panel.grid.major.x = element_line(color = "gray95",
                                           linewidth = 0.5),
        axis.text.x = element_text(face = "bold",
                                    size = 10,
                                    color = "black"))

# Creating histogram with KDE of intercept with mean and confidence interval

boot_p_intercept <- data.frame(boot_intercepts)
boot_my_breaks <- round(c(boot_quantile_intercept[1],
                        boot_mean_intercept,
                        boot_quantile_intercept[2]), 1)

ggplot(data = boot_p_intercept,
       aes(x = boot_intercepts)) +
  labs(title = "Bootstrap Empirical Distribution of the Intercept",
       subtitle = "Red line: Bootstrap Mean | Blue lines: 95% Confidence Interval",
       x = "Intercept Coefficient Value (b)",
       y = "Density") +
  geom_histogram(aes(y=after_stat(density)), # Histogram of values
                bins = 50,
                fill = "lightblue",
                color = "black") +
  geom_density(color = "black", # KDE of values
              linewidth = 1.2) +
  geom_vline(xintercept = c(boot_mean_intercept, # Mean and confidence interval lines
                           boot_quantile_intercept[1],
                           boot_quantile_intercept[2]),
            color = c("red", "darkblue", "darkblue"),
            linetype = c("dashed", "dotted", "dotted"),
            linewidth = 1) +
  scale_x_continuous(breaks = boot_my_breaks, # Mean and confidence interval values
                    labels = scales::label_comma()) +
  theme_minimal() +
  theme(panel.grid.major.x = element_line(color = "gray95",
                                           linewidth = 0.5),
        axis.text.x = element_text(face = "bold",
                                    size = 10,
                                    color = "black"))

```

```

        color = c("red", "darkblue", "darkblue"),
        linetype = c("dashed", "dotted", "dotted"),
        linewidth = 1) +
scale_x_continuous(breaks = boot_my_breaks, # Mean and confidence interval values
                    labels = scales::label_comma()) +
theme_minimal() +
theme(panel.grid.major.x = element_line(color = "gray95",
                                          linewidth = 0.5), #
axis.text.x = element_text(face = "bold",
                             size = 10,
                             color = "black"))

# Constructing Monte Carlo simulation

# Calculating simulation parameters from values of base OLS

mc_true_a <- unname(coef(model)[2])
mc_true_b <- unname(coef(model)[1])
mc_std <- summary(model)$sigma

X_mc <- rnorm(n, mean = mean(data$population), sd = sd(data$population))
mc_slopes <- numeric(rep)
mc_intercepts <- numeric(rep)

# Monte Carlo simulation

for(i in 1:rep)
{
  mc_eps <- rnorm(n, mean = 0, sd = mc_std)
  Y_mc <- mc_true_a * X_mc + mc_true_b + mc_eps
  model_mc <- lm(Y_mc ~ X_mc)

  mc_slopes[i] <- coef(model_mc)[2]
  mc_intercepts[i] <- coef(model_mc)[1]
}

# Calculating the results

mc_mean_slope <- mean(mc_slopes)
mc_mean_intercept <- mean(mc_intercepts)

mc_quantile_slope <- quantile(mc_slopes, probs = c(0.025, 0.975))
mc_quantile_intercept <- quantile(mc_intercepts, probs = c(0.025, 0.975))

# Creating histogram with KDE of slope coefficient with real value and confidence interval

mc_p_slope <- data.frame(mc_slopes)
mc_my_breaks <- as.numeric(round(c(mc_quantile_slope[1],
                                mc_true_a,
                                mc_quantile_slope[2]), 5))

ggplot(data = mc_p_slope,

```

```

    aes(x = mc_slopes)) +
labs(title = "Monte Carlo Simulation of the Slope Coefficient",
     subtitle = "Red line: True Value | Blue lines: 95% Confidence Interval",
     x = "Estimated Slope Value",
     y = "Density") +
geom_histogram(aes(y=after_stat(density)), # Histogram of values
               bins = 50,
               fill = "lightblue",
               color = "black") +
geom_density(color = "black", # KDE of values
             linewidth = 1.2) +
geom_vline(xintercept = c(mc_true_a, # True value and confidence interval lines
                          mc_quantile_slope[1],
                          mc_quantile_slope[2]),
           color = c("red", "darkblue", "darkblue"),
           linetype = c("dashed", "dotted", "dotted"),
           linewidth = 1) +
  scale_x_continuous(breaks = mc_my_breaks) + # True value and confidence interval values
theme_minimal() +
theme(panel.grid.major.x = element_line(color = "gray95",
                                         linewidth = 0.5),
      axis.text.x = element_text(face = "bold",
                                  size = 10,
                                  color = "black"))

# Creating histogram with KDE of intercept coefficient with real value and confidence interval

mc_p_intercept <- data.frame(mc_intercepts)
mc_my_breaks <- as.numeric(c(mc_quantile_intercept[1],
                             mc_true_b,
                             mc_quantile_intercept[2]))

ggplot(data = mc_p_intercept,
       aes(x = mc_intercepts)) +
labs(title = "Monte Carlo Simulation of the Intercept Coefficient",
     subtitle = "Red line: True Value | Blue lines: 95% Confidence Interval",
     x = "Estimated Intercept Value",
     y = "Density") +
geom_histogram(aes(y=after_stat(density)), # Histogram of values
               bins = 50,
               fill = "lightblue",
               color = "black") +
geom_density(color = "black", # KDE of values
             linewidth = 1.2) +
geom_vline(xintercept = c(mc_true_b, # True value and confidence interval lines
                          mc_quantile_intercept[1],
                          mc_quantile_intercept[2]),
           color = c("red", "darkblue", "darkblue"),
           linetype = c("dashed", "dotted", "dotted"),
           linewidth = 1) +
  scale_x_continuous(breaks = mc_my_breaks) + # True value and confidence interval values
theme_minimal() +

```

```

theme(panel.grid.major.x = element_line(color = "gray95",
                                          linewidth = 0.5),
      axis.text.x = element_text(face = "bold",
                                  size = 10,
                                  color = "black"))

# Conducting prediction for 2015 data

prediction <- predict(model, newdata = test_data)

# Calculating simulation results

errors <- test_data$marriages - prediction
mae <- mean(abs(errors))
rmse <- sqrt(mean(errors^2))
mape <- mean(abs(errors) / test_data$marriages) * 100

t.test(errors, mu = 0)

# Conducting FD simulation

model_diff <- lm(d_marriages ~ d_population + 0)

summary(model_diff)

```